

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – Data Interpretation

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO4 – Precision	Assessment:	Assessment Tool 1 – Data Interpretation
Weightage:	35% of CIA		SLOT B
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	AI & ML	Date:	17-08-2026

Code of Conduct

I G.V.S.HEMANTH/192425186(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: G.V.S.HEMANTH

Instructions

- Read each data set, table, semantic representation, or scenario carefully before answering.
- Analyze the given information and provide logical, evidence-based interpretations.
- Show all intermediate reasoning, semantic analysis steps, predicate representations, or calculations wherever applicable.
- Answers should be concise, structured, and technically accurate.
- Support your conclusions using the data provided in the question.
- Draw diagrams, semantic networks, parse trees, or logical representations wherever relevant.
- Avoid copying definitions alone; focus on analyzing the given scenario and interpreting the data.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze semantic representations, identify action–entity relationships, detect interpretation errors, and evaluate meaning representation in chatbot systems.	Q1
First-Order Predicate Calculus & Representing Linguistically Relevant Concepts	Construct predicate representations, apply logical inference rules, derive conclusions, and evaluate reasoning in smart manufacturing environments.	Q2
Lexemes and Their Senses & Word Sense Disambiguation	Analyze contextual clues, determine correct word senses, evaluate ambiguity resolution, and improve semantic understanding in search systems.	Q3
Syntax-Driven Semantic Analysis	Analyze syntactic structures, assign semantic roles, evaluate parsing accuracy, and improve semantic interpretation in healthcare NLP applications.	Q4

Annexure A – Data Interpretation

1. Semantic Representation in Customer Support Chatbots.

A telecom company collected customer chatbot interactions. The semantic analyzer generated the following meaning representations.

Customer Query	Semantic Representation
"Activate international roaming for my number."	ACTIVATE(Roaming, Customer)
"Deactivate caller tune service."	DEACTIVATE(CallerTune, Customer)
"Check my data balance."	QUERY(DataBalance, Customer)
"Enable 5G service."	ACTIVATE(5GService, Customer)

Additional chatbot logs show:

Query ID	Actual User Intent	Predicted Intent
Q1	Activate Roaming	Activate Roaming
Q2	Deactivate Caller Tune	Activate Caller Tune
Q3	Check Data Balance	Query Data Balance
Q4	Enable 5G Service	Activate 5G Service

Tasks:

1. Analyze the semantic representations and identify the action-object relationships.
 2. Determine which query contains semantic interpretation errors.
 3. Evaluate how meaning representation affects chatbot decision accuracy.
 4. Recommend improvements to enhance semantic understanding in telecom customer support systems.
- ### 2. First-Order Predicate Calculus for Smart Manufacturing.
- An Industry 4.0 manufacturing plant monitors machines using semantic rules.
Production Data:

Machine	Status
M1	Active
M2	Active
M3	Maintenance
M4	Active

Predicate Rules:

- $\text{Active}(x) \rightarrow \text{Producing}(x)$
- $\text{Produces}(x,y) \wedge \text{Active}(x) \rightarrow \text{Available}(y)$
- $\text{Maintenance}(x) \rightarrow \neg \text{Producing}(x)$

Tasks:

1. Represent the production data using First-Order Predicate Calculus.
 2. Analyze which products are currently available.
 3. Infer whether Gear production is affected by maintenance activities.
 4. Evaluate the effectiveness of predicate logic in industrial decision-support systems.
- ### 3. Word Sense Disambiguation in E-Commerce Search Engines.
- An e-commerce company observed the following search logs.

User Query	Possible Sense 1	Possible Sense 2
"Apple accessories"	Fruit	Technology Brand
"Mouse wireless"	Animal	Computer Device
"Java tutorial"	Island	Programming Language
"Python course"	Snake	Programming Language

Search Context Data:

Query	Clicked Result
Apple accessories	iPhone Charger
Mouse wireless	Bluetooth Mouse
Java tutorial	Coding Lessons
Python course	Software Development Training

Tasks:

1. Analyze the contextual information and determine the correct word sense for each query.
2. Identify the semantic cues used for disambiguation.
3. Evaluate the impact of incorrect sense selection on search engine performance.
4. Suggest an industrial-scale WSD strategy for improving recommendation accuracy.

4. Syntax-Driven Semantic Analysis in Healthcare Information Systems.

A healthcare NLP system processes medical reports.

Parsed Sentences:

Sentence	Parse Structure
Doctor prescribed medicine to patient.	Subject-Verb-Object
Patient reported severe headache.	Subject-Verb-Object
Nurse monitored patient continuously.	Subject-Verb-Object
Medicine reduced blood pressure.	Subject-Verb-Object

Semantic Roles Generated:

Entity	Role
Doctor	Agent
Medicine	Instrument
Patient	Recipient
Headache	Symptom

Tasks:

1. Analyze how syntactic structures contribute to semantic role assignment.
2. Determine whether the assigned semantic roles are appropriate.
3. Evaluate potential errors that could arise from incorrect parsing.
4. Recommend methods to improve syntax-driven semantic analysis in healthcare applications.

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Q. No. / Criteria Level	Understanding of Data	Analysis	Application of Concepts	Conclusion	Clarity	Sub. Total	Total %
1							
2							
3							
4							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Assessment Tool 1 – Data Interpretation

Q1. Semantic Representation in Customer Support Chatbots

1. Action–Object Relationships

The semantic representations follow the structure:

ACTION(Object, Entity)

Customer Query	Semantic Representation	Action	Object
Activate international roaming for my number	ACTIVATE (Roaming, Customer)	Activate	Roaming
Deactivate caller tune service	DEACTIVATE (CallerTune, Customer)	Deactivate	Caller Tune
Check my data balance	QUERY (DataBalance, Customer)	Query	Data Balance
Enable 5G service	ACTIVATE (5GService, Customer)	Activate	5G Service

The representation correctly identifies the customer's intended action and the relevant telecom service.

2. Query with Semantic Interpretation Error

Q2 contains the semantic interpretation error.

- Actual intent: **Deactivate Caller Tune**
- Predicted intent: **Activate Caller Tune**

The semantic representation should be:

DEACTIVATE (CallerTune, Customer)

but the chatbot effectively interprets it as:

ACTIVATE (CallerTune, Customer)

The error is in the **action/predicate**, while the object, Caller Tune, is correctly identified.

3. Effect on Chatbot Decision Accuracy

Meaning representation directly affects the action selected by the chatbot.

- Q1: Correct interpretation → correct activation.
- Q2: Incorrect interpretation → potentially activates a service that the customer wants to deactivate.
- Q3: Correct interpretation → retrieves the data balance.
- Q4: Correct interpretation → activates 5G service.

Therefore, **3 out of 4 queries are correctly interpreted**, giving an observed semantic decision accuracy of:

$$\left[\begin{array}{l} \text{Accuracy} = \frac{3}{4} \times 100 = 75\% \end{array} \right]$$

The Q2 error is particularly serious because confusing **activate** and **deactivate** can result in an action opposite to the customer's request.

4. Recommended Improvements

The chatbot can be improved by:

1. Using intent classification that distinguishes opposite actions such as **activate/deactivate**, **enable/disable**, and **subscribe/unsubscribe**.
2. Using contextual and syntactic analysis to identify negation and action words.
3. Representing entities and actions separately in a structured semantic frame.
4. Adding confidence scores and confirmation for potentially irreversible service changes.
5. Training with contrasting examples such as:
 - "Activate roaming."
 - "Deactivate roaming."
 - "Do not activate roaming."
6. Using domain-specific telecom terminology and continuously evaluating the system using confusion matrices and intent-level accuracy.

Conclusion: Accurate semantic representation is essential because even when the correct entity is detected, an incorrect action predicate can cause the chatbot to perform the wrong operation.

Q2. First-Order Predicate Calculus for Smart Manufacturing

1. Representation of Production Data

The machine status can be represented using First-Order Predicate Calculus as:

- Active (M1)
- Active (M2)
- Maintenance (M3)
- Active (M4)

The rules are:

$$\left[\begin{array}{l} \forall x; \text{Active}(x) \rightarrow \text{Producing}(x) \end{array} \right]$$

$$\left[\begin{array}{l} \forall x \forall y; \text{Produces}(x,y) \wedge \text{Active}(x) \rightarrow \text{Available}(y) \end{array} \right]$$

]

[
 $\forall x; \text{Maintenance}(x) \rightarrow \neg \text{Producing}(x)$
]

From the first rule:

- $\text{Active}(M1) \rightarrow \text{Producing}(M1)$
- $\text{Active}(M2) \rightarrow \text{Producing}(M2)$
- $\text{Active}(M4) \rightarrow \text{Producing}(M4)$

From the maintenance rule:

- $\text{Maintenance}(M3) \rightarrow \neg \text{Producing}(M3)$

Thus, M1, M2 and M4 are producing, while M3 is not producing.

2. Products Currently Available

The second rule states:

[
 $\text{Produces}(x,y) \wedge \text{Active}(x) \rightarrow \text{Available}(y)$
]

Therefore, a product is known to be available **only when there is evidence that an active machine produces that product.**

The supplied data do not specify any $\text{Produces}(\text{machine}, \text{product})$ facts. Hence, no specific product can be logically identified as available from the given information alone.

For example, if the plant contained:

$\text{Produces}(M1, \text{Gear})$

then because:

$\text{Active}(M1)$

we could infer:

$\text{Available}(\text{Gear})$

3. Is Gear Production Affected by Maintenance?

The data establish:

$\text{Maintenance}(M3) \rightarrow \neg \text{Producing}(M3)$

However, there is **no fact stating that M3 produces Gear.**

Therefore, it is not logically valid to conclude that Gear production is affected by M3's maintenance from the given data alone.

If an additional fact were provided:

`Produces (M3, Gear)`

then maintenance of M3 would imply that M3 is not producing, and Gear production through M3 would be affected.

Thus, the correct conclusion is:

Gear production may be affected if Gear depends on M3, but this dependency is not specified in the given data.

4. Effectiveness of Predicate Logic

Predicate logic is effective for industrial decision-support because it:

- Represents machine states precisely.
- Encodes operational rules formally.
- Supports logical inference from machine status.
- Can identify conditions such as production stoppage.
- Reduces ambiguity in automated decision-making.

However, its effectiveness depends on the completeness and accuracy of the knowledge base. Missing relationships, such as which machine produces Gear, prevent the system from making more specific conclusions.

Conclusion: First-Order Predicate Calculus provides a precise foundation for reasoning about manufacturing systems, but accurate decisions require complete machine-product relationships in addition to machine-status data.

Q3. Word Sense Disambiguation in E-Commerce Search Engines

1. Correct Word Sense

Query	Context / Clicked Result	Correct Sense
Apple accessories	iPhone Charger	Technology/Apple brand
Mouse wireless	Bluetooth Mouse	Computer device
Java tutorial	Coding Lessons	Programming language
Python course	Software Development Training	Programming language

The clicked results provide strong evidence for the intended meaning.

2. Semantic Cues for Disambiguation

The system can use several contextual cues:

Apple accessories

- "Accessories" is strongly associated with electronic products.
- "iPhone Charger" confirms the technology/brand interpretation.
- Fruit-related accessories are unlikely in this context.

Mouse wireless

- "Wireless" is strongly associated with electronic devices.
- "Bluetooth Mouse" confirms the computer-device sense.

Java tutorial

- "Tutorial" combined with "coding lessons" indicates programming.
- Java as an island does not fit the context.

Python course

- "Course" and "software development training" indicate programming.
- The snake interpretation is inconsistent with the clicked result.

3. Impact of Incorrect Sense Selection

Incorrect word-sense selection can:

- Return irrelevant search results.
- Reduce click-through rate.
- Decrease user satisfaction.
- Lower recommendation accuracy.
- Increase search abandonment.
- Reduce conversion rates in e-commerce.
- Cause incorrect personalization.

For example, interpreting **"Mouse wireless"** as an animal could return wildlife-related content instead of computer accessories. Similarly, interpreting **"Python course"** as a snake could produce zoology content rather than programming courses.

Therefore, WSD has a direct effect on both **search relevance and commercial performance**.

4. Industrial-Scale WSD Strategy

An industrial e-commerce WSD system should combine:

1. **Contextual embeddings** from transformer-based language models.
2. **Query-level context**, including surrounding words such as "wireless", "tutorial", and "course".
3. **Product catalog information**, including categories, attributes, and product descriptions.
4. **User behavior**, such as clicks, purchases, and browsing history.
5. **Knowledge graphs** linking terms to products, categories, and entities.
6. **Click-through and conversion feedback** to continuously improve predictions.
7. **Candidate sense ranking**, where possible meanings are scored using context.
8. **Real-time model updating** to handle new products, brands, and terminology.

Conclusion: The clicked results demonstrate that contextual information can strongly disambiguate polysemous words. A large-scale WSD system should combine linguistic context with catalog and behavioral information to improve search and recommendation accuracy.

Q4. Syntax-Driven Semantic Analysis in Healthcare Information Systems

1. Contribution of Syntax to Semantic Role Assignment

Syntactic structure helps determine **who performed an action, what was affected, and who received or experienced it.**

For example:

"Doctor prescribed medicine to patient."

Syntactic structure:

Doctor → prescribed → medicine → to patient

This can be interpreted as:

- Doctor → Agent
- Medicine → Theme
- Patient → Recipient

Similarly:

"Patient reported severe headache."

- Patient → Experiencer
- Severe headache → Symptom

Syntax therefore provides important evidence for assigning semantic roles.

2. Appropriateness of Assigned Semantic Roles

Some roles are appropriate, but one important assignment is incorrect.

Entity	Given Role	Evaluation	Better Role
Doctor	Agent	Appropriate	Agent
Medicine	Instrument	Incorrect	Theme
Patient	Recipient	Appropriate for "prescribed medicine to patient"	Recipient
Headache	Symptom	Appropriate	Symptom

Medicine should normally be classified as the Theme, because it is the entity being prescribed. It is not an instrument used by the doctor to perform the prescribing action.

Additional sentence-level roles include:

- Patient in "Patient reported..." → **Experiencer**
- Nurse in "Nurse monitored..." → **Agent**
- Patient in "Nurse monitored patient..." → **Patient/Theme**
- Medicine in "Medicine reduced blood pressure..." → **Agent/Cause**
- Blood pressure → **Theme/affected entity**

3. Errors Caused by Incorrect Parsing

Incorrect syntactic parsing can cause serious semantic errors in healthcare NLP.

Examples include:

- Assigning the wrong patient to a medical treatment.
- Confusing the doctor with the recipient of medication.
- Misidentifying symptoms.
- Incorrectly extracting medication information.
- Reversing cause-and-effect relationships.
- Producing incorrect clinical summaries.
- Affecting clinical decision-support systems.

For example, if a parser incorrectly interprets "**Doctor prescribed medicine to patient**", the system could associate the medicine with the wrong entity or assign the patient an incorrect role.

4. Methods to Improve Syntax-Driven Semantic Analysis

Healthcare NLP systems should use:

1. **Dependency parsing** to identify grammatical relationships.
2. **Semantic Role Labeling (SRL)** to assign Agent, Patient, Theme, Experiencer, and other roles.
3. **Healthcare-specific language models** trained on medical text.
4. **Medical ontologies and terminology databases** to recognize diseases, drugs, symptoms, and procedures.
5. **Negation and uncertainty detection**, such as distinguishing "patient has headache" from "patient does not have headache."
6. **Coreference resolution** to determine which patient or condition a pronoun refers to.
7. **Human-validated clinical datasets** for training and evaluation.
8. **Confidence-based review** for ambiguous or clinically important interpretations.

Conclusion: Syntax provides the structural foundation for semantic interpretation, but syntax alone is insufficient. Combining dependency parsing, semantic role labeling, medical terminology, negation detection, and domain-specific language models can produce more reliable healthcare information extraction.

Overall Conclusion

The four datasets demonstrate that **semantic interpretation depends on accurate representation of meaning, logical relationships, contextual word senses, and syntactic structure.**

- Semantic representations help chatbots map user requests to correct actions.
- First-Order Predicate Calculus enables formal reasoning in manufacturing systems.
- Word Sense Disambiguation improves search relevance by selecting meanings from context.
- Syntax-driven semantic analysis enables healthcare systems to identify clinically meaningful roles and relationships.

In all four applications, errors in semantic interpretation can lead to incorrect decisions. Therefore, combining **symbolic representations, contextual information, syntactic analysis, domain knowledge, and machine-learning techniques** is essential for building reliable NLP systems.